a

ROC curves, IxIDN-Matched (n = 4,364; 50% positive)

True positive rate

False positive rate

AUROC
DisAtlas (full) 0.832
DisAtlas (benchmark diseases held out) 0.784
Claude (chat) 0.692
Claude (science agent) 0.685
Phylo Biomni 0.627
TxGNN 0.566
Gemini Pro 0.497

b

Ranking metrics (dot = value, bar = 95% CI, diseases resampled)

AUPRC
F1 at k = #pos.
Precision, top 10%
NPV, bottom 10%

DisAtlas (full): 0.842, 0.760, 0.956, 0.867
DisAtlas (benchmark diseases held out): 0.789, 0.719, 0.913, 0.851
Claude (chat): 0.739, 0.628, 0.933, 0.635
Claude (science agent): 0.708, 0.640, 0.860, 0.638
Phylo Biomni: 0.664, 0.596, 0.830, 0.500
TxGNN: 0.606, 0.541, 0.773, 0.530
Gemini Pro: 0.536, 0.497, 0.631, 0.489
random

c

Run-to-run agreement of pair scores

one point = one pair of runs; no box under 5 pairs; TxGNN (single run) not shown

Spearman ρ between two runs

DisAtlas (full)
DisAtlas (benchmark diseases held out)
Claude (chat)
Claude (science agent)
Phylo Biomni
Gemini Pro

d

Overlap of top-10% pairs (Jaccard; 436 of 4,364 per method)

	DisAtlas (full)	DisAtlas (benchmark diseases held out)	Claude (chat)	Claude (science agent)	Phylo Biomni	TxGNN	Gemini Pro
DisAtlas (full)	1.00	0.47	0.27	0.20	0.17	0.12	0.12
DisAtlas (benchmark diseases held out)	0.47	1.00	0.23	0.16	0.16	0.12	0.10
Claude (chat)	0.27	0.23	1.00	0.51	0.26	0.12	0.14
Claude (science agent)	0.20	0.16	0.51	1.00	0.26	0.09	0.15
Phylo Biomni	0.17	0.16	0.26	0.26	1.00	0.06	0.19
TxGNN	0.12	0.12	0.12	0.09	0.06	1.00	0.02
Gemini Pro	0.12	0.10	0.14	0.15	0.19	0.02	1.00

Jaccard index